

Surender Singh

AI & Data Science Engineer | Machine Learning | Computer Vision | NLP | Full-Stack AI Development

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Professional Summary

Results-driven AI and Data Science Engineer with hands-on expertise in building and deploying end-to-end machine learning systems spanning Computer Vision, Natural Language Processing, and Predictive Analytics. Skilled in bridging research-grade deep learning models with production-ready full-stack applications using modern MLOps and web technologies. Completed 2 industry internships, built 6 deployed AI/ML projects, and earned 8 professional certifications. Proficient in PyTorch, TensorFlow, scikit-learn, HuggingFace Transformers, YOLOv8, OpenCV, Flask, and the MERN stack. Seeking roles in Machine Learning Engineering, Data Science, or AI Development.

Education

Bachelor of Technology (B.Tech) — Artificial Intelligence & Data Science | CGPA: 7.58

Modern Institute of Technology and Research Center (MITRC), Alwar, Rajasthan | 2023 – 2027 (Expected)

Technical Skills

Programming Languages: Python (Expert), JavaScript, C++, Java, SQL, HTML, CSS

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, LSTM, CNN, Gradient Boosting, Feature Engineering, Predictive Modeling, EDA

Computer Vision: YOLOv8, OpenCV, MediaPipe, PyAutoGUI, Real-Time Inference

Natural Language Processing: HuggingFace Transformers, T5, LSTM with Attention, Text Summarization, Tokenization

Data Analytics: Pandas, NumPy, Matplotlib, Seaborn, Risk Profiling, Fraud Detection, Data Storytelling

Full-Stack & Deployment: Flask, React.js, Node.js, Express.js, MongoDB, MySQL, JWT, REST API, Chart.js

MLOps & DevOps: Git, GitHub, Docker, Model Deployment, API Integration

Generative AI: LLM, RAG, Agentic AI, Prompt Engineering, GenAI Workflows

Professional Experience

Data Science Virtual Intern | CodSoft

Remote | April 2026

- Completed a 4-week structured virtual internship focused on Data Science workflows including exploratory data analysis (EDA), feature engineering, and predictive modeling.
- Applied Python-based machine learning pipelines (scikit-learn, Pandas, NumPy) on real-world datasets to extract actionable business insights and build classification and regression models.
- Performed data visualization using Matplotlib and Seaborn to communicate findings clearly to non-technical stakeholders.

Full Stack Web Development Intern | WSA (Web Stack Academy)

Remote | March 2026 – April 2026

- Developed a full-stack web application following SDLC practices using the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Built RESTful APIs with Node.js and Express.js; implemented JWT-based authentication and connected a React front-end to a MongoDB database.
- Gained production-level exposure to HTML, CSS, JavaScript, and React component architecture.

Projects

TaskFlow

MERN Stack, Node.js, MongoDB, Chart.js, REST API

- Built an intelligent full-stack productivity system with a custom Intelligence Engine for automated task prioritization and behavioral recommendations using rule-based ML logic.
- Integrated Chart.js dashboards for real-time productivity metric tracking and task completion visualization across user sessions.

PlayStore Pulse

Python, Flask, Gradient Boosting, React.js, REST API, scikit-learn

- Developed an analytics platform processing 10,000+ Google Play Store app records with interactive dashboards for ratings, installs, and category insights.
- Trained and deployed a Gradient Boosting rating predictor via a Flask REST API — full end-to-end ML deployment from model training to live API consumption.

Emotion Classifier

YOLOv8, PyTorch, CNN, OpenCV, Computer Vision, Deep Learning

- Built a real-time facial emotion recognition system combining YOLOv8 face detection with a custom-trained CNN (PyTorch) classifying across 7 emotion categories with live webcam inference at production-ready latency.

Hand Gesture Virtual Mouse

MediaPipe, OpenCV, PyAutoGUI, Computer Vision, HCI

- Engineered a touchless human-computer interaction (HCI) system using MediaPipe hand landmark tracking to map 21-point finger gestures to OS-level controls (mouse clicks, drag, scroll, zoom) — eliminating physical input devices entirely.

Dual-Model Text Summarizer

HuggingFace Transformers, T5, LSTM, Attention Mechanism, PyTorch, NLP

- Designed a comparative NLP tool implementing a custom LSTM with Attention mechanism (trained from scratch) alongside a fine-tuned HuggingFace T5 transformer for side-by-side abstractive summarization benchmarking and performance evaluation.

Certifications

- Tata — GenAI Powered Data Analytics Job Simulation | Forage | April 2026
- CodSoft — Data Science Virtual Internship | CodSoft | April 2026
- Artificial Intelligence Journey: Beginner to Pro | Udemy | April 2026
- Machine Learning & AI Foundations Course | Udemy | April 2026
- Certified Master in Artificial General Intelligence | Udemy School of AI | April 2026
- Python Mastery: 100 Days, 100 Projects | Udemy School of AI | February 2026
- Google Generative AI Leader Exam: Complete Prep Guide | Udemy | January 2026
- Ultimate DevSecOps Bootcamp | Udemy School of DevOps | February 2026
- Agentic AI Playbook: Complete Guide for Tech Leaders | Udemy School of DevOps | February 2026

Achievements

- 1st Prize — College Hackathon: Outperformed competing teams in a competitive technical challenge, demonstrating rapid prototyping, system design, and problem-solving under time constraints.